

EMILY OUK

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EDUCATION

UNIVERSITY OF CALIFORNIA, SAN DIEGO

Bachelor of Science in Aerospace Engineering | EMPOWER Scholar |

Relevant Coursework: Thermodynamics, Statics, Fluids | Planned Coursework: Solid Mechanics, Aerodynamics, Signals & Systems

La Jolla, CA

Graduating June 2028

EXPERIENCE

ROCKET PROPULSION LABORATORY (UC San Diego)

La Jolla, CA

Propulsion & Avionics Engineer | Project Phoenix

September 2025 - Current

- Developed the avionics integration, programmed microcontrollers in Arduino to interface with GPS, altitude, and pressure sensors
- Collaborated with the development and instrumentation of the propulsion test stand for 13-foot liquid methalox rocket

Aerospace Engineer | Project Daedalus

- Modeled and optimized flight trajectory using OpenRocket and MATLAB, measuring CG and improving stability margins by 80%
- Integrated a Smiski payload into a rocket mission and successfully recovered the figurine post-launch, achieving mission objectives
- Launched a G-class rocket integrating parachute recovery systems and airframe systems, implemented PDR/CDR feedback
- Modeled and optimized custom rocket fins through SolidWorks flow simulations and iterative 3D-printed prototyping

TRITON ASTRONAUTICS RESEARCH SYSTEMS (TARS)

La Jolla, CA

Director of Astronautics and Payload

September 2025 - March 2025

- Launched a high-altitude balloon payload to 80,000 ft and successfully recovered the payload and avionics system post-flight
- Supported launch-day operations by assisting with helium system components & payload preparation during balloon deployment

WOMEN OF AERONAUTICS AND ASTRONAUTICS

La Jolla, CA

Development Chair

May 2026 - Present

- Led organization and execution of project-based initiatives, including L1 model rocketry assembly and certification events
- Coordinated and planned technical workshops and laboratory tours while empowering marginalized genders in aerospace fields

NASA COMMUNITY COLLEGE AEROSPACE SCHOLARS (NCAS)

Hamptons, VA (Hybrid)

Project Manager | Surface Operations Specialist | Communications Specialist

August 2024 - July 2025

- Served as Project Manager guiding team dynamics and refining a 25-page proposal into a presentation for NASA staff, and peers
- Led a team of eight and developed a solution to use autonomous electrical Vertical Take Off and Landing (eVTOL) drones to deliver groceries to socioeconomically disadvantaged citizens living in rural communities
- Collaborated with a team of eight to deliver a high-scoring mission proposal, earning a perfect score from NASA evaluators

NASA L'SPACE NPWEE

Tempe, AZ (Remote)

Lead System Engineer (Team Role)

May 2024 - August 2024

- Served as Lead Scientist Engineer on a cross-functional team that designed a 17-page proposal for a fire detection software system tailored to microgravity environments, conducted research, and collaborated with a large team of 12-15 members
- Contributed as a panelist reviewer for two peer proposals, providing detailed technical feedback on design feasibility, methodology, and mission objectives while gaining insight into evaluation criteria used in aerospace funding decisions

PROJECTS

DAEDALUS AVIONICS SYSTEM | Arduino IDE, Circuiting, Wiring, Soldering

- Designed and implemented an embedded avionics system by integrating sensors: BMP388, BNO055 IMU, GPS module, and microSD storage with an Arduino Nano using Arduino-based embedded software for flight data acquisition and flight analysis
- Built and prototyped circuitry on a protoboard, including sensor integration, wiring, and soldered electrical connections
- Conducted full-system flight testing, successfully launching and recovering avionics hardware with all onboard sensors remaining

AUTONOMOUS LINE FOLLOWING ROBOT | Onshape, ESP32, Arduino IDE, PID Control

- Assembled an autonomous line-tracking robot, integrating an ESP32 microcontroller, custom CAD chassis, & power systems
- Implemented PID control and PWM motor regulation using photoresistor feedback to optimize steering and autonomous navigation

PAYLOAD STRUCTURE | Solidworks, 3D Printing, Propeller-Motor Thrust Stand

- Designed a functional structure in SolidWorks to hold two 130.5g payloads at fixed, separated positions on a pivoting thrust beam
- Performed force and torque analysis utilizing a thrust coefficient of 0.134, successfully lifted the payload into horizontal equilibrium

SOCCER BALL KICK SIMULATION | MATLAB

- Developed a MATLAB simulation to model the motion of a soccer ball during a free kick, analyzing gravitational & Magnus forces
- Generated 3D surface-rendered plots tracking the ball's trajectory relative to the field boundaries, the goal, and a dynamic goalkeeper and defenders, optimizing the script to fully execute all simulations in under five minutes

SKILLS

Technical skills: C++, LaTeX, MatLab, SolidWorks, 3D Printing, OpenRocket, Soldering, Arduino IDE, KiCad, Breadboard Design

Currently learning: ANSYS (FEA/Thermal), Computer Numerical Control Machining (CNC)

Soft Skills: Technical writing, Time management, Problem-Solving, Adaptability, Organized, Communication

Hi, I'm Dani Ouk. I'm an aspiring Aerospace Engineering student at the University of California, San Diego. I earned three Associate Degrees in Mathematics, Physics, and Physical Sciences at Long Beach City College.

My passion for space exploration, technological innovation, and solving challenges within the aerospace sector has driven me to pursue this field. During my time at community college, I have been involved in a variety of extracurricular activities, most notably NASA NCAS and NASA L'SPACE.

As I transition into UCSD, I intend to apply the knowledge I obtain from my upper-division courses into hands-on aerospace related projects, research, or internships.